Pinocchio-EP and ANSI/ASHRAE Standard 140-2023: Conformance Testing and Engineering Practice

ERGO Energy LLP — Pinocchio-EP Engineering
Release `1.0.0+e2c8ebf` · EnergyPlus 26.1.0-6f2e40d102
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***Abstract*—Pinocchio-EP is a web-based building energy modelling application that authors EnergyPlus models through a guided interface and runs an unmodified EnergyPlus 26.1 engine. ANSI/ASHRAE Standard 140-2023 treats the interface and the engine together as the software under test. We therefore tested the released Pinocchio-EP build itself:**

- **scope: every test section of the standard, Sections 6–13, including Addendum a's hourly acceptance criteria and Addendum b's empirical ETNA tests;**
- **inputs: ASHRAE's official accompanying weather files and output templates;**
- **build path: every model built through Pinocchio-EP's own IDF writer.**

All five graded test groups of Annex A3 pass: 136 of 136 range cases are inside the acceptance ranges. For the four constant-coefficient ETNA cases, modelled heater energy agrees with the measurements to within 0.03–0.33 %. We describe the engineering practices behind this result: unmodified-engine provenance, no silent input transformations, version stamping, standard-format reporting, regression gates, and bit-level parity with openstudio-standards. We also list the limits that remain.

***Index Terms*—ASHRAE Standard 140, BESTEST, building energy simulation, software validation, EnergyPlus, empirical validation, ETNA**

I. Introduction

Standard 140 [1] is the method of test that energy codes and incentive programmes use to qualify simulation software. It is required by ASHRAE 90.1-2022 §12.4.1.4 and Appendices C and G [5], by green-building rating systems built on 90.1, and by the US DOE §179D qualified-software process [11]. Its Annex A3 contains a no-proxy rule: results obtained through one user interface cannot document another interface on the same engine. A tool built on EnergyPlus must therefore be tested as itself.

This report presents the results for the released Pinocchio-EP build and the practices that support them. All inputs, outputs, filled ASHRAE workbooks and modeller reports are archived with the release.

II. The Standard and Its Addenda

Class I tests cover:

- weather processing (Section 6);
- thermal fabric (Section 7);
- ground coupling (Section 8);
- cooling, heating and air-side equipment (Sections 9–11);
- Addendum b's empirical ETNA tests (Section 13).

Section 12 holds the Class II tests for simplified tools.

Annex A3 grades five test groups (Table I). Addendum a [2] adds a time-series criterion to the thermal-fabric groups: the hourly loads on 1 February and 14 July must exceed a Pearson correlation of 0.879 against the reference median curve, and must stay below a case-specific RMSE limit. Addendum b [3] adds six empirical cases from the guarded ETNA twin test cells [10]. Their comparison against measured heater energy is informative only; there is no pass/fail.

III. Architecture and Engineering Practices

A. Engine Provenance

The server writes the IDF produced in the browser verbatim and invokes the official NREL EnergyPlus 26.1.0 Linux release [6]. The build is pinned by hash `6f2e40d102`, and the container build fails if the binary or its version-transition programs are missing. The desktop test engine is the identical build. The engine is never patched; engine defects are reported (Section VII) rather than worked around in code.

B. One Model, No Silent Transformations

Every Pinocchio-EP step edits one structured IDF model, and the IDF the user downloads is exactly the IDF that runs. This release removed three silent behaviours found in the first audit:

- imported files had their time step capped at 4 per hour;
- a missing weather file was replaced by a default climate;
- the site-exposure field never reached the model.

Numerical settings now pass through unchanged, and the Review step displays them. A requested but missing weather file is an error. Every run reports the weather file it used.

C. Faithful Inputs

The interface can express the standard's inputs exactly:

- layered glazing with spectral-average pane properties (Fig. 4b);
- materials with exact conductivity, density, specific heat and absorptances;
- constant convection coefficients (Fig. 4d);
- full-wall overhangs and full-height fins;
- other-side-coefficient boundaries;
- Kiva ground coupling [8];
- DX performance maps fitted inside the application;
- any EnergyPlus object, through an IDD-validated "add object" dialog (Fig. 4e).

Weather in TMY3, TMY2, TMY or WYEC2 format is converted to EPW on import [9]. The converter is validated hour-by-hour against ASHRAE's own EPW files.

D. Validation Profile

An opt-in "Standard 140 / validation" profile applies the standard's test conditions, each documented with its clause:

- no daylight saving time or holidays (5.1.1.1);
- no latent removal by ideal loads (7.2.1.13);
- interior solar distribution (7.2.1.12);
- the specified terrain (7.2.1.9.1).

Normal projects keep realistic defaults. This separation prevents test-specific assumptions from leaking into design work.

E. Traceability and Reporting

Each build carries a version identifier, `1.0.0+<git sha>`. It is shown in the interface and written into every exported IDF, as Annex A2 block A requires. The application produces the Annex A2 modeller report (`S140outNotes`, blocks A–G) and fills ASHRAE's official output workbooks in place (Fig. 5d).

F. Verification Discipline

- **Tests:** 992 automated front-end tests and 210 back-end tests pass on the tested build.
- **Regression gate:** it reruns Cases 600 and 900 on the real engine and checks them against Annex A3.
- **GUI–agent equivalence:** every interface field is also exposed to the AI agent through a registry, and a test proves that GUI and agent produce identical models.
- **OpenStudio parity:** the HVAC emitter reproduces openstudio-standards [7] exactly. All 117 parity cases match, with 0 text or order differences across 32 805 objects and a maximum difference of zero across 9 014 040 hourly meter values.
- **Independence from reference answers:** no input or code was adjusted to match reference results, as 5.1.1.10 requires; each disclosed modelling choice is recorded in block C.

TABLE I
ANNEX A3 GRADED TEST GROUPS

Test group	In range	Needed	Result
Thermal fabric, low mass	34/34	30	**Pass**
Thermal fabric, high mass	28/28	25	**Pass**
Cooling equipment (Sec. 9)	14/14	12	**Pass**
Heating equipment (Sec. 10)	6/6	5	**Pass**
Air-side HVAC (Sec. 11)	54/54	48	**Pass**
Total	**136/136**		**5/5**

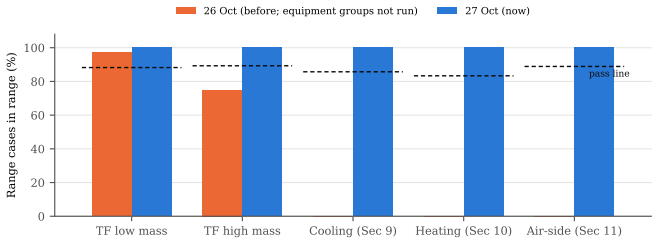


Fig. 1. Share of range cases in range per group, before the fixes (first audit) and for the tested release.

IV. Test Method

Every case is a Pinocchio-EP project or imported IDF, exported by the same `exportIdf` routine as the interface's Run button and executed on the pinned engine. The Section 7 cases use only data the interface writes. Inputs come from ASHRAE's official 140-2023 accompanying files [4] and the Addendum b ETNA files. Annex A3 scoring is implemented in the application (`bestest.js`).

V. Results

A. Thermal Fabric (Section 7)

All 52 cases ran with no severe errors: 600–695, 900–995, 960, six free-float cases, and the in-depth cases 195–470, 800 and 810. Base results:

- Case 600: heating 4.301 MWh, sensible cooling 6.015 MWh;
- Case 900: heating 1.661 MWh, sensible cooling 2.506 MWh.

All 22 Addendum a day curves pass.

B. HVAC Equipment (Sections 9–11)

- **Cooling, CE100–CE200:** within 0.3–1.8 % of the quasi-analytical total energy.
- **Cooling, CE300:** 35 039 kWh/yr, against a range of 33 497–37 023.
- **Heating, HE100:** furnace input 77.756 GJ against an analytical 77.713 GJ.
- **Heating, HE210:** 52.694 GJ, against a range of 49.86–55.11.
- **Air-side:** all 54 ranged quantities pass.

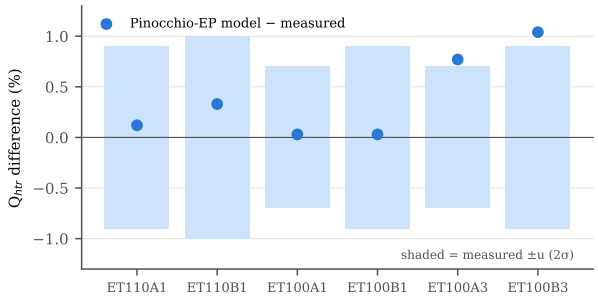


Fig. 2. ETNA: modelled minus measured heater energy; the shaded band is the measured uncertainty $\pm u$ (2σ).

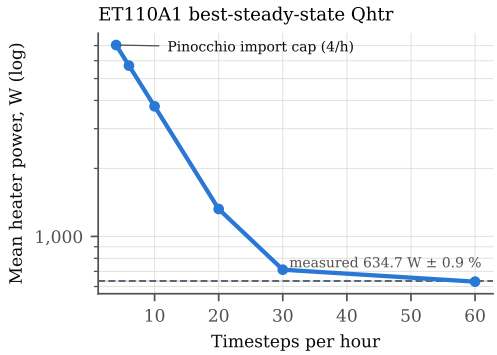


Fig. 3. First audit: forcing an ETNA input to 4 steps/h changed heater power tenfold. The release keeps the modeller's time step.

C. Informative Sections

- **Section 6:** the engine reproduces the weather file to 10^{-5}, and 55 of 60 annual incident-solar values fall inside the reference-program range.
- **Section 8:** 16 of 17 cases ran with Kiva, within -7.4 to $+1.2\,\%$ of the verified numerical models.
- **Section 12:** 38 of 58 results fall inside the informative example ranges.

D. Empirical Validation (Section 13)

- **Constant-coefficient cases:** modelled heater energy is within $+0.03$ to $+0.33\,\%$ of measured (Fig. 2). Every best-steady-state hour lies inside the measurement uncertainty band.
- **Automatic-convection cases (A3/B3):** $+0.77$ and $+1.04\,\%$, using the measured supply airflow as specified.
- **Time step:** for a model built from the specification, the result changes by less than $0.1\,\%$ between 4 and 60 steps per hour. The earlier import-path error in Fig. 3 is now removed.

VI. User-Interface Evidence

Figs. 4 and 5 show the released interface features that carry the standard's inputs and outputs. Table II maps each requirement to its implementation.

TABLE II
Standard Requirement to Implementation

Requirement	Pinocchio-EP implementation
Engine + UI tested together (A3)	every case through `exportIdf`
Time convention, no DST (5.1.1.1)	validation profile
Consistent numerics (5.1.1.4, 5.1.1.9)	time step never altered; shown in Review
Documented equivalent inputs (5.1.1.5)	S140outNotes block C
Weather formats (Sec. 6, A1)	TMY3/TMY2/TMY/WYEC2 converters
Envelope inputs (Sec. 7)	layered glazing, exact materials, shading
Constant coefficients (7.2.3)	convection panel
Ground coupling (Sec. 8)	Kiva panel
Equipment maps (Sec. 9–10)	map fitting, lookup part-load curves
Air systems (Sec. 11)	Studio: FC, SZ, CV and VAV reheat
Measured boundaries (Sec. 13)	other-side coefficients, schedule import
Outputs (7.3, A2)	output picker, peaks, report export
Version identifier (A2 block A)	build stamp in UI and IDF

VII. Limitations and External Defects

- **Section 8:** GC10a cannot be represented in Kiva, which is two-dimensional and has no ground-temperature output points.
- **Section 12:** the results show a low bias in the window-related terms; its cause is under investigation.

Defects found in EnergyPlus 26.1, all reported and none patched:

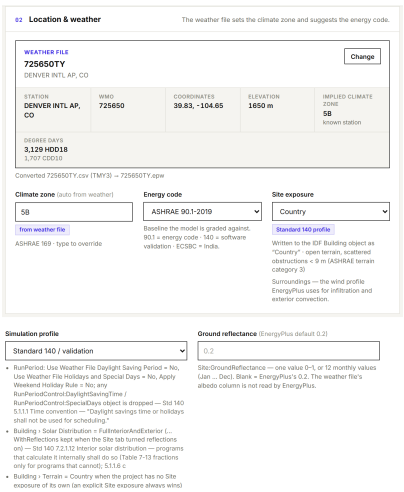
- the FisherPedersen ceiling-diffuser keys are fatal in `MultipleSurface`;
- the economizer lockout is ignored on load-controlled unitary systems;
- economizer-first staging crashes the engine;
- the EPW albedo column is ignored;
- wind direction is interpolated straight through north.

Defects found in the standard's own files, all handled and documented:

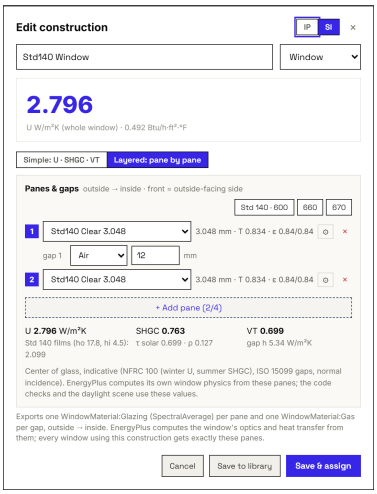
- an equation-of-time typo in A1.5;
- an inconsistent WD400 header;
- a TMY local-standard-time offset;
- a duplicated WYEC2 hour stamp.

VIII. Conclusion

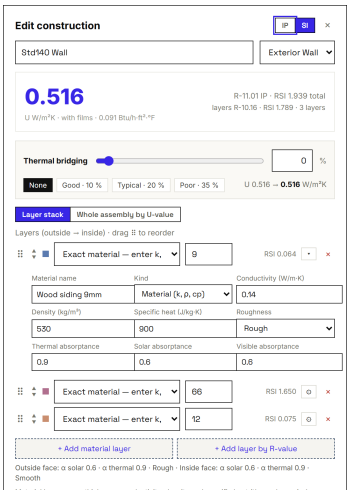
The tested release of Pinocchio-EP passes every graded test group of ANSI/ASHRAE Standard 140-2023, with Addendum a's time-series criteria, using ASHRAE's official files. It also reproduces Addendum b's measured heater energy to within the measurement uncertainty. The result rests on documented practices: a pinned, unmodified engine; inputs that are never silently altered; standard-format reporting; versioned builds; automated regression against Annex A3. The full suite was generated on build `1.0.0+b7b2379`. The deployed release



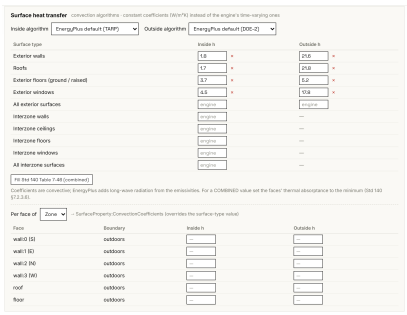
(a) Project step: Std 140 profile, site exposure, TMY3 upload converted to EPW, ground reflectance.



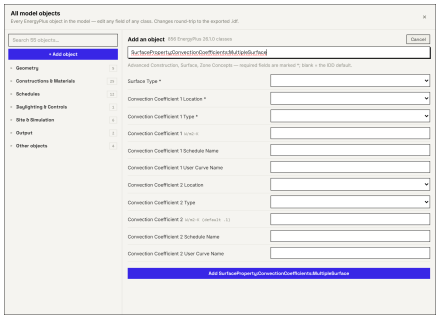
(b) Layered glazing: Std 140 clear double pane; U and SHGC computed.



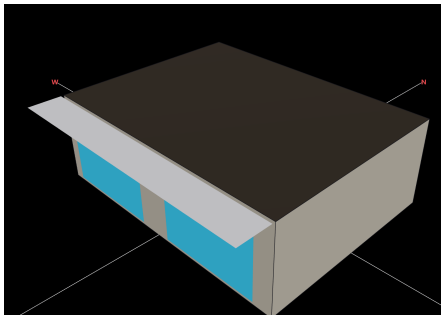
(c) Exact material layers (Case 600 wall).



(d) Surface heat transfer filled with Table 7-46 coefficients.



(e) IDD-validated "add object" dialog (all 856 classes).



(f) Case 610 full-wall overhang in 3-D.

Fig. 4. Input-side interface features of the tested release that carry Standard 140 inputs (Sections 6–7).

`1.0.0+e2c8ebf` changes only a Review display card; the on-engine Annex A3 regression gate and all 992 tests were re-run on it. Future releases are re-tested by the same harness.

References

[1] *Method of Test for Evaluating Building Performance Simulation Software*, ANSI/ASHRAE Standard 140-2023, ASHRAE, Peachtree Corners, GA, USA, 2023.

[2] *Addendum a to ANSI/ASHRAE Standard 140-2023*, ANSI/ASHRAE/IBPSA, Aug. 2025.

[3] *Addendum b to ANSI/ASHRAE Standard 140-2023*, ANSI/ASHRAE/IBPSA, Apr. 2026.

[4] ASHRAE, "Standard 140-2023 accompanying files," [Online]. Available: https://data.ashrae.org/standard140/

[5] *Energy Standard for Sites and Buildings Except Low-Rise Residential Buildings*, ANSI/ASHRAE/IES Standard 90.1-2022, ASHRAE, 2022.

[6] U.S. Department of Energy, *EnergyPlus Version 26.1.0 Documentation*, 2026. [Online]. Available: https://energyplus.net

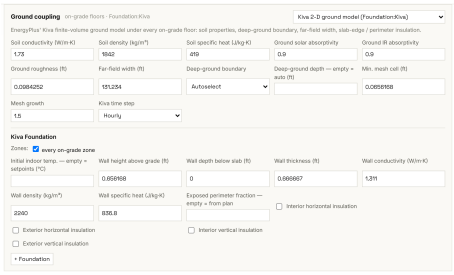
[7] National Renewable Energy Laboratory, "openstudio-standards," v0.8.5. [Online]. Available: https://github.com/NREL/openstudio-standards

[8] N. Kruis and M. Krarti, "Kiva™: A numerical framework for quickly modeling ground-coupled heat transfer," *J. Building Performance Simulation*, vol. 10, no. 5–6, 2017.

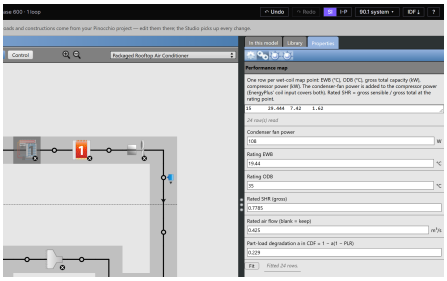
[9] S. Wilcox and W. Marion, "Users manual for TMY3 data sets," NREL, Golden, CO, USA, Tech. Rep. NREL/TP-581-43156, 2008.

[10] ETNA test facility (EDF R&D), as specified in Addendum b to ASHRAE Standard 140-2023, Section 13.

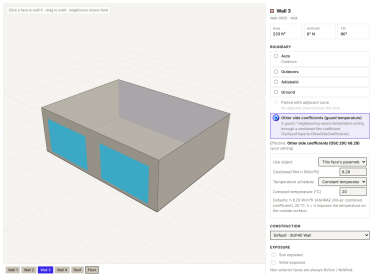
[11] U.S. Department of Energy, "Qualified software for calculating commercial building tax deductions," [Online]. Available: https://www.energy.gov/eere/buildings/qualified-software-calculating-commercial-building-tax-deductions



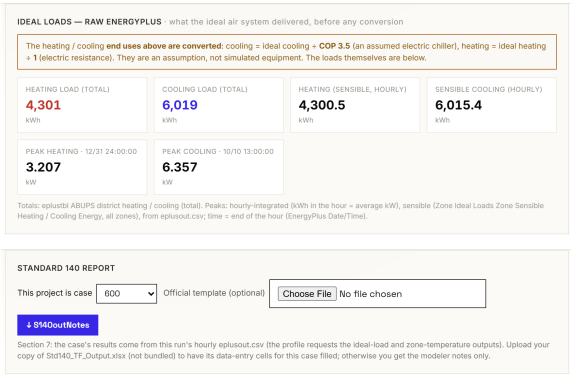
(a) Kiva ground coupling (Section 8).



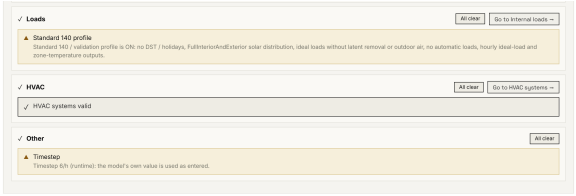
(b) DX coil performance-map fitting, CE100 map (Section 9).



(c) Other-side-coefficient boundary (Section 13 guards).



(d) Results: raw ideal loads with timestamped peaks and the Std 140 report export (Case 600 run).



(e) Review: profile notice; the model's own time step is used unaltered.

Fig. 5. System, boundary, output and validation features (Sections 8–13, Annex A2).